

ALFAAC26837

## Toluene

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                                             |                                                                                                                                                                                                                                                                    |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>产品说明:</b><br><b>Product Description:</b> | <b>甲苯</b><br><b>Toluene</b>                                                                                                                                                                                                                                        |
| <b>Cat No. :</b>                            | <b>C26837</b>                                                                                                                                                                                                                                                      |
| <b>Synonyms</b>                             | Tol; Methylbenzene                                                                                                                                                                                                                                                 |
| <b>CAS No</b>                               | 108-88-3                                                                                                                                                                                                                                                           |
| <b>Molecular Formula</b>                    | C7 H8                                                                                                                                                                                                                                                              |
| <b>Supplier</b>                             | Avocado Research Chemicals Ltd.<br>(Part of Thermo Fisher Scientific)<br>Shore Road, Heysham<br>Lancashire, LA3 2XY,<br>United Kingdom<br>Office Tel: +44 (0) 1524 850506<br>Office Fax: +44 (0) 1524 850608                                                       |
| <b>Emergency Telephone Number</b>           | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| <b>E-mail address</b>                       | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| <b>Recommended Use</b>                      | Laboratory chemicals.                                                                                                                                                                                                                                              |
| <b>Uses advised against</b>                 | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                                                                                                                                                                                                                                                                                                                                             |                                |                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------|
| <b>Physical State</b><br>Liquid                                                                                                                                                                                                                                                                                                                                             | <b>Appearance</b><br>Colorless | <b>Odor</b><br>aromatic |
| <b>Emergency Overview</b><br>Highly flammable liquid and vapor. May be fatal if swallowed and enters airways. Causes skin irritation. Suspected of damaging fertility or the unborn child. May cause drowsiness and dizziness. Toxic to aquatic life. Harmful to aquatic life with long lasting effects. May cause damage to organs through prolonged or repeated exposure. |                                |                         |

#### Classification of the substance or mixture

|                                                      |            |
|------------------------------------------------------|------------|
| Flammable liquids.                                   | Category 2 |
| Aspiration Toxicity                                  | Category 1 |
| Skin Corrosion/Irritation                            | Category 2 |
| Reproductive Toxicity                                | Category 2 |
| Specific target organ toxicity - (single exposure)   | Category 3 |
| Specific target organ toxicity - (repeated exposure) | Category 2 |
| Acute aquatic toxicity                               | Category 2 |
| Chronic aquatic toxicity                             | Category 3 |

#### Label Elements

## Toluene



## Signal Word

## Danger

## Hazard Statements

H225 - Highly flammable liquid and vapor  
H304 - May be fatal if swallowed and enters airways  
H315 - Causes skin irritation  
H336 - May cause drowsiness or dizziness  
H361 - Suspected of damaging fertility or the unborn child  
H373 - May cause damage to organs through prolonged or repeated exposure  
H401 - Toxic to aquatic life  
H412 - Harmful to aquatic life with long lasting effects

## Precautionary Statements

## Prevention

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear eye protection/ face protection

## Response

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P331 - Do NOT induce vomiting  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse

## Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Physical and Chemical Hazards

Vapors may cause flash fire or explosion. Highly flammable.

## Health Hazards

Aspiration hazard if swallowed - can enter lungs and cause damage. Causes skin irritation. Suspected of damaging fertility or the unborn child. May cause drowsiness or dizziness. May cause damage to organs through prolonged or repeated exposure.

## Environmental hazards

Toxic to aquatic life. Harmful to aquatic life with long lasting effects. Is not likely mobile in the environment due its low water solubility. The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Spillage unlikely to penetrate soil. The product is insoluble and floats on water.

## Other Hazards

Toxic to terrestrial vertebrates. This product does not contain any known or suspected endocrine disruptors.

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## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component | CAS No   | Weight % |
|-----------|----------|----------|
| Toluene   | 108-88-3 | <=100    |

## SECTION 4. FIRST AID MEASURES

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. Risk of serious damage to the lungs (by aspiration).

**Ingestion**

Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward.

**Most important symptoms and effects**

Difficulty in breathing. Causes central nervous system depression: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically. Smallest quantities reaching the lungs through swallowing or subsequent vomiting may result in lung edema or pneumonia. Symptoms may be delayed.

## SECTION 5. FIRE-FIGHTING MEASURES

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

**Extinguishing media which must not be used for safety reasons**

Do not use water jetstream.

**Specific Hazards Arising from the Chemical**

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6. ACCIDENTAL RELEASE MEASURES

**Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

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**Environmental Precautions**

Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Ensure adequate ventilation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Storage**

Keep containers tightly closed in a dry, cool and well-ventilated place. Flammables area. Keep away from heat, sparks and flame.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component | China                                                            | Taiwan                                     | Thailand                                          | Hong Kong                                 |
|-----------|------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------|-------------------------------------------|
| Toluene   | TWA: 50 mg/m <sup>3</sup><br>STEL: 100 mg/m <sup>3</sup><br>Skin | TWA: 100 ppm<br>TWA: 376 mg/m <sup>3</sup> | Ceiling: 300 ppm<br>STEL: 500 ppm<br>TWA: 200 ppm | TWA: 50 ppm<br>TWA: 188 mg/m <sup>3</sup> |

| Component | ACGIH TLV   | OSHA PEL                                                                                                                                                               | NIOSH                                                                                                       | The United Kingdom                                                                                                        | European Union                                                                                                                |
|-----------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Toluene   | TWA: 20 ppm | (Vacated) TWA: 100 ppm<br>(Vacated) TWA: 375 mg/m <sup>3</sup><br>Ceiling: 300 ppm<br>(Vacated) STEL: 150 ppm<br>(Vacated) STEL: 560 mg/m <sup>3</sup><br>TWA: 200 ppm | IDLH: 500 ppm<br>TWA: 100 ppm<br>TWA: 375 mg/m <sup>3</sup><br>STEL: 150 ppm<br>STEL: 560 mg/m <sup>3</sup> | STEL: 100 ppm 15 min<br>STEL: 384 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 191 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 50 ppm (8hr)<br>TWA: 192 mg/m <sup>3</sup> (8hr)<br>STEL: 100 ppm (15min)<br>STEL: 384 mg/m <sup>3</sup> (15min)<br>Skin |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Monitoring methods**

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents. MDHS70 General methods for sampling airborne gases and vapours MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

**Exposure Controls****Engineering Measures**

Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof

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electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

## Personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard       | Glove comments                                                                                                               |
|----------------|-------------------|-----------------|-------------------|------------------------------------------------------------------------------------------------------------------------------|
| Viton (R)      | < 240 minutes     | 0.30 mm         | Level 4<br>EN 374 | Permeation rate 68 µg/cm <sup>2</sup> /min<br>As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Viton (R)      | > 480 minutes     | 0.70 mm         |                   |                                                                                                                              |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Valve filtering: EN405; or, Half mask: EN140; plus filter, EN 141  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                 |                           |                                          |
|---------------------------------|---------------------------|------------------------------------------|
| <b>Appearance</b>               | Colorless                 |                                          |
| <b>Physical State</b>           | Liquid                    |                                          |
| <b>Odor</b>                     | aromatic                  |                                          |
| <b>Odor Threshold</b>           | 1.74 ppm                  |                                          |
| <b>pH</b>                       | No information available  |                                          |
| <b>Melting Point/Range</b>      | -95 °C / -139 °F          |                                          |
| <b>Softening Point</b>          | No data available         |                                          |
| <b>Boiling Point/Range</b>      | 111 °C / 231.8 °F         | @ 760 mmHg                               |
| <b>Flash Point</b>              | 4 °C / 39.2 °F            | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>         | 2.4 (Butyl acetate = 1.0) |                                          |
| <b>Flammability (solid,gas)</b> | Not applicable            | Liquid                                   |
| <b>Explosion Limits</b>         | <b>Lower</b> 1.2 vol%     |                                          |

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|                                                |                                      |                                             |
|------------------------------------------------|--------------------------------------|---------------------------------------------|
| <b>Vapor Pressure</b>                          | Upper 7 vol%                         |                                             |
| <b>Vapor Density</b>                           | 29 mbar @ 20 °C                      |                                             |
| <b>Specific Gravity / Density</b>              | 3.1                                  | (Air = 1.0)                                 |
| <b>Bulk Density</b>                            | 0.866                                |                                             |
| <b>Water Solubility</b>                        | Not applicable                       | Liquid                                      |
| <b>Solubility in other solvents</b>            | practically insoluble 0.5 g/L @ 20°C |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> | No information available             |                                             |
| <b>Component</b>                               | <b>log Pow</b>                       |                                             |
| Toluene                                        | 2.73                                 |                                             |
| <b>Autoignition Temperature</b>                | 535 °C / 995 °F                      |                                             |
| <b>Decomposition Temperature</b>               | No data available                    |                                             |
| <b>Viscosity</b>                               | 0.6 mPa.s @ 20 °C                    |                                             |
| <b>Explosive Properties</b>                    | Not explosive                        | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | Not oxidising                        |                                             |
| <b>Molecular Formula</b>                       | C7 H8                                |                                             |
| <b>Molecular Weight</b>                        | 92.14                                |                                             |

## SECTION 10. STABILITY AND REACTIVITY

|                                         |                                                                                                       |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Stable under normal conditions.                                                                       |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                         |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                              |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Materials to avoid</b>               | Strong oxidizing agents. Strong acids. Strong bases. Halogenated compounds.                           |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).                                              |

## SECTION 11. TOXICOLOGICAL INFORMATION

## Product Information

## (a) acute toxicity;

| Component | LD50 Oral            | LD50 Dermal            | LC50 Inhalation       |
|-----------|----------------------|------------------------|-----------------------|
| Toluene   | > 5000 mg/kg ( Rat ) | 12000 mg/kg ( Rabbit ) | 26700 ppm ( Rat ) 1 h |

## (b) skin corrosion/irritation;

|                               |                    |
|-------------------------------|--------------------|
| <b>Test method</b>            | Category 2         |
| <b>Test species</b>           | OECD 404           |
| <b>Observational endpoint</b> | rabbit             |
|                               | Irritating to skin |

(c) serious eye damage/irritation; Based on available data, the classification criteria are not met

## (d) respiratory or skin sensitization;

|                    |                                                                  |
|--------------------|------------------------------------------------------------------|
| <b>Respiratory</b> | Based on available data, the classification criteria are not met |
| <b>Skin</b>        | Based on available data, the classification criteria are not met |

## (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met  
Not mutagenic in AMES Test

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|                                                                                               |                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (f) carcinogenicity;                                                                          | Based on available data, the classification criteria are not met<br>There are no known carcinogenic chemicals in this product                                                                           |
| (g) reproductive toxicity;<br>Reproductive Effects<br>Developmental Effects<br>Teratogenicity | Category 2<br>Experiments have shown reproductive toxicity effects on laboratory animals.<br>Developmental effects have occurred in experimental animals.<br>Possible risk of harm to the unborn child. |
| (h) STOT-single exposure;<br><br>Results / Target organs                                      | Category 3<br><br>Central nervous system (CNS)                                                                                                                                                          |
| (i) STOT-repeated exposure;<br><br>Target Organs                                              | Category 2<br><br>Liver, Kidney, Central nervous system (CNS), Blood, spleen, Neuropsychological effects, Eyes, Ears.                                                                                   |
| (j) aspiration hazard;                                                                        | Category 1                                                                                                                                                                                              |
| Symptoms / effects, both acute and delayed                                                    | Causes central nervous system depression: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting                                           |

## SECTION 12. ECOLOGICAL INFORMATION

**Ecotoxicity effects** The product contains following substances which are hazardous for the environment.  
Contains a substance which is: Toxic to aquatic organisms.

| Component | Freshwater Fish                                                                                              | Water Flea                                                                                         | Freshwater Algae                                                                                                                 | Microtox                |
|-----------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| Toluene   | 50-70 mg/L LC50 96 h<br>5-7 mg/L LC50 96 h<br>15-19 mg/L LC50 96 h<br>28 mg/L LC50 96 h<br>12 mg/L LC50 96 h | EC50: = 11.5 mg/L, 48h<br>(Daphnia magna)<br>EC50: 5.46 - 9.83 mg/L,<br>48h Static (Daphnia magna) | EC50: = 12.5 mg/L, 72h static<br>(Pseudokirchneriella subcapitata)<br>EC50: > 433 mg/L, 96h<br>(Pseudokirchneriella subcapitata) | EC50 = 19.7 mg/L 30 min |

**Persistence and Degradability** Readily biodegradable  
**Persistence** Persistence is unlikely.

| Component                     | Degradability |
|-------------------------------|---------------|
| Toluene<br>108-88-3 ( <=100 ) | 86% (20d)     |

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Toluene   | 2.73    | 90                            |

**Mobility in soil** The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces Spillage unlikely to penetrate soil The product is insoluble and floats on water Is not likely mobile in the environment due its low water solubility

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

## SAFETY DATA SHEET

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**Persistent Organic Pollutant  
Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused  
Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

UN-No UN1294  
Proper Shipping Name TOLUENE  
Hazard Class 3  
Packing Group II

**IMDG/IMO**

UN-No UN1294  
Proper Shipping Name TOLUENE  
Hazard Class 3  
Packing Group II

**IATA**

UN-No UN1294  
Proper Shipping Name TOLUENE  
Hazard Class 3  
Packing Group II

**Special Precautions for User** No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|-----------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Toluene   | X                                                   | X                                       | X    | X     | 203-625-9 | X    | X   | X     | X    | X    | X    | KE-33936 |

**National Regulations**



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## SECTION 16. OTHER INFORMATION

**Prepared By** Health, Safety and Environmental Department  
**Creation Date** 11-Jun-2009  
**Revision Date** 13-May-2024  
**Revision Summary** New emergency telephone response service provider.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**